

# Crude Transfer Pump Failure

## Field Manual — Setup and Troubleshooting Reference

**Use case definition.** A crude transfer pump moves stabilized or semi-stabilized crude between tanks, separators, heaters, and export systems. Normal operation is defined by stable suction and discharge pressure, required flow, acceptable motor load, bearing temperature, vibration, and seal condition. Abnormal behavior includes loss of flow, pressure deviation, overheating, leakage, trips, or mechanical noise. Causes commonly include suction restriction, cavitation, blocked discharge, poor lubrication, seal or bearing damage, misalignment, loss of prime, and electrical faults. Troubleshooting is required to prevent loss of containment, equipment damage, and interruption of transfer operations.

**Manual setup and use.** Before use, record the pump tag, pump type, rotation direction, rated flow, OEM alarm/trip limits, lubrication type, seal arrangement, and approved isolation procedure. Use the site permit, LOTO, hazardous-area requirements, and the applicable OEM/site alarm or trip limits. Values below are training/reference operating values, not universal shutdown limits.

**Safety note.** Do not open, loosen, or repair pressurized crude equipment. For leakage, isolate and depressurize using the approved site procedure before maintenance. Treat hot crude, rotating equipment, and hydrocarbon vapour as hazards.

## Quick-use workflow

Use this sequence for every event: establish a safe operating state; capture the initial numerical readings; compare each value with the stated reference; apply the trigger-based action; correct the identified component or process path; restore operation gradually; and confirm the expected numerical result. If an OEM/site alarm or trip limit differs from a training value, the OEM/site limit governs.

## Scenario 1: Bearing overheating

### Field Condition

bearing temperature 94 °C (normal 65–75 °C); vibration 6.8 mm/s RMS (normal 2–4); flow 185 m<sup>3</sup>/h; discharge 5.4 barg

### Problem

Bearing temperature is above the 90 °C training trigger and vibration is elevated, indicating a possible lubrication, alignment, or bearing condition.

### What Can Be Done

1. Confirm the local RTD against a calibrated handheld reading; if both exceed 90 °C, reduce load and prepare a controlled shutdown.
2. Compare oil level with the maker marks; if below minimum or oil is dark/contaminated, isolate and correct lubrication under LOTO.
3. Measure vibration at both bearing housings; investigate values above the applicable OEM/site alarm limit and compare axial versus radial readings.
4. After isolation, check coupling alignment, foundation bolts, pipe strain, and bearing roughness; repair the identified mechanical defect.
5. Restart gradually and accept operation only when temperature trends toward 65–75 °C, vibration toward 2–4 mm/s RMS, and flow remains 180–200 m<sup>3</sup>/h.

### Expected Result

Bearing temperature 65–75 °C, vibration 2–4 mm/s RMS, and flow 180–200 m<sup>3</sup>/h.

## Scenario 2: Mechanical seal leakage

### Field Condition

seal leakage 0.35 L/min (normal less than 0.05); suction 2.0 barg; discharge 5.1 barg; seal-pot pressure 0.8 barg

### Problem

Visible leakage is seven times the reference rate and presents a loss-of-containment risk.

## What Can Be Done

1. Keep personnel outside the vapour path and verify the leak rate from a safe location; if detector reading exceeds the site action level, stop and isolate.
2. Confirm suction and discharge valves are in the operating lineup and that discharge pressure is not above 5.5 barg.
3. Check seal flush pressure against the approved range; if it is below 1.0 barg reference, restore the flush source only from a safe, approved location.
4. If leakage remains above 0.10 L/min, execute the approved shutdown, isolation, depressurization, and drain procedure; do not tighten a pressurized gland.
5. Replace or repair the seal, then leak-test at the approved test pressure and restart only when leakage is below 0.05 L/min.

## Expected Result

Leakage below 0.05 L/min with stable seal-flush pressure and no hydrocarbon detector response.

## Scenario 3: Cavitation

### Field Condition

suction pressure 0.35 barg (normal 1.8–2.2); discharge 5.0 barg; flow 132 m<sup>3</sup>/h (normal 180–200); vibration 7.2 mm/s RMS

### Problem

Low suction pressure, reduced flow, and broadband vibration are consistent with insufficient NPSH or suction restriction.

## What Can Be Done

1. If suction pressure falls below 1.0 barg or vibration exceeds the site alarm limit, unload or stop the pump using the approved procedure.
2. Confirm tank level is above the minimum operating level and compare the level transmitter with the local gauge; investigate a difference above 5 %.
3. Check suction valve position and strainer differential pressure; a rise from 0.2 to 0.8 bar indicates likely restriction.
4. Restore tank level or clean the isolated strainer as authorized; do not remove a strainer cover while pressurized.
5. Restart at reduced speed/load and confirm suction 1.8–2.2 barg, flow 180–200 m<sup>3</sup>/h, and vibration 2–4 mm/s RMS.

## Expected Result

Stable suction 1.8–2.2 barg, flow 180–200 m<sup>3</sup>/h, and vibration 2–4 mm/s RMS.

## Scenario 4: Low suction pressure

### Field Condition

suction 0.62 barg (normal 1.8–2.2); tank level 19 % (normal 35–70); flow 148 m<sup>3</sup>/h; motor current 78 A

### Problem

The pump is being starved by low source level or an upstream restriction.

## What Can Be Done

1. Confirm suction pressure with the local gauge; if below 1.0 barg, reduce speed/load and follow the controlled shutdown criterion in the operating procedure.
2. Compare tank level transmitter and sight indication; if level is below 25 %, stop transfer or switch to the approved source tank.
3. Verify the suction valve is at least 95 % open and inspect the strainer differential pressure from the indicators.
4. If differential pressure is at least 0.6 bar, isolate and clean the strainer under LOTO.
5. Restore the source level and restart only after suction reaches 1.8–2.2 barg and flow returns to 180–200 m<sup>3</sup>/h.

## Expected Result

Suction 1.8–2.2 barg, source level above 35 %, and flow 180–200 m<sup>3</sup>/h.

## Scenario 5: Blocked suction strainer

### Field Condition

strainer differential pressure 0.92 bar (normal 0.2–0.3); suction 1.1 barg; discharge 5.2 barg; flow 142 m<sup>3</sup>/h

### Problem

The high differential pressure indicates a restricted suction path.

### What Can Be Done

1. Reduce pump load if suction approaches 1.0 barg and prevent operation below the site minimum suction pressure.
2. Verify both strainer pressure taps and compare with a calibrated differential gauge; accept the restriction as confirmed at 0.8 bar or higher.
3. Transfer to the standby strainer if the lineup permits, then isolate, drain, and depressurize the fouled strainer.
4. Remove debris, inspect the basket for collapse, and replace damaged mesh before reassembly.
5. Restore the strainer gradually and verify differential pressure 0.2–0.3 bar, suction 1.8–2.2 barg, and flow 180–200 m<sup>3</sup>/h.

### Expected Result

Strainer differential pressure 0.2–0.3 bar and flow 180–200 m<sup>3</sup>/h.

## Scenario 6: High discharge pressure

### Field Condition

discharge 8.4 barg (normal 5.0–5.5); suction 2.1 barg; flow 96 m<sup>3</sup>/h; motor current 86 A (normal 68–75)

### Problem

A blocked discharge path or closed downstream valve is causing high head and motor load.

### What Can Be Done

1. If discharge exceeds 7.0 barg or current exceeds 80 A, unload/stop according to the approved pump procedure.
2. Compare the local discharge gauge with the transmitter; treat a difference above 0.3 bar as an instrumentation fault requiring verification.
3. Confirm the discharge valve and downstream isolation valves are open to the intended destination; do not force a seized valve.
4. Check downstream filter/flowline differential pressure; a value above the normal 0.3 bar may indicate restriction.
5. Correct the lineup or isolate the blocked section, then restart and confirm discharge 5.0–5.5 barg, flow 180–200 m<sup>3</sup>/h, and current 68–75 A.

### Expected Result

Discharge 5.0–5.5 barg, flow 180–200 m<sup>3</sup>/h, current 68–75 A.

## Scenario 7: Low discharge pressure

### Field Condition

discharge 2.7 barg (normal 5.0–5.5); suction 2.0 barg; flow 78 m<sup>3</sup>/h; speed 1,480 RPM

### Problem

Low head with low flow suggests loss of prime, reverse rotation, internal wear, or a suction-side problem.

### What Can Be Done

1. Verify discharge pressure locally; if below 3.5 barg with the pump commanded on, place the pump in a controlled stop.
2. Confirm rotation against the pump arrow during a safe bump test; correct phase sequence only by an authorized electrician.
3. Check casing vent/priming status and suction valve alignment; restore prime using the approved procedure.
4. If prime is present, compare speed with the 1,480 RPM reference and inspect impeller/clearance during isolation.
5. Restart and confirm discharge 5.0–5.5 barg and flow 180–200 m<sup>3</sup>/h.

## Expected Result

Discharge 5.0–5.5 barg and flow 180–200 m<sup>3</sup>/h at approximately 1,480 RPM.

## Scenario 8: Reduced flow

### Field Condition

flow 118 m<sup>3</sup>/h (normal 180–200); suction 1.7 barg; discharge 5.3 barg; current 61 A (normal 68–75)

### Problem

Flow is low despite near-normal pressures, suggesting a partially closed valve, plugged meter, or worn impeller.

### What Can Be Done

1. Validate flow using the local meter or timed tank volume; treat a difference above 5 % as an instrument issue.
2. Verify suction and discharge valve positions are at least 95 % open where required by the operating lineup.
3. Compare pump differential pressure with the curve at 1,480 RPM; a 5.3 barg discharge with low current can indicate reduced hydraulic loading.
4. Inspect the flow meter impulse path and downstream filter under an approved isolation.
5. If hydraulics are clear, inspect impeller wear and clearance, then restart only after flow reaches 180–200 m<sup>3</sup>/h.

## Expected Result

Flow 180–200 m<sup>3</sup>/h with current 68–75 A and stable pressures.

## Scenario 9: Motor overload

### Field Condition

motor current 92 A (rated reference 80 A); discharge 5.8 barg; flow 214 m<sup>3</sup>/h; bearing 72 °C

### Problem

The motor is overloaded by excessive flow/head or mechanical drag.

### What Can Be Done

1. If current exceeds the motor nameplate or site overload setting, reduce speed/load and follow the trip-prevention procedure.
2. Confirm current on all three phases; imbalance above 10 % indicates an electrical issue requiring authorized electrical testing.
3. Reduce flow toward 180–200 m<sup>3</sup>/h using the approved control method; do not throttle the suction valve.
4. Check discharge pressure and coupling/bearing condition after isolation if current remains above 80 A.
5. Resume only when current is 68–75 A, flow 180–200 m<sup>3</sup>/h, and temperature remains 65–75 °C.

## Expected Result

Current 68–75 A and flow 180–200 m<sup>3</sup>/h without overload alarm.

## Scenario 10: Motor trip

### Field Condition

pump stopped on overload at 08:42; last current 101 A; voltage 398 V (normal 400–415); discharge 5.9 barg

### Problem

The motor protection trip may result from overload, low voltage, phase imbalance, or a mechanically seized pump.

### What Can Be Done

1. Do not reset repeatedly; record the trip value and apply the approved electrical isolation before opening panels or touching the coupling.
2. Verify all phase voltages are within the site tolerance around 400–415 V and phase imbalance is below 2 %.
3. With the pump isolated, confirm shaft turns freely and inspect for bearing seizure or blocked discharge.
4. Correct the mechanical or electrical cause and test protection with an authorized electrician.

- Restart unloaded where possible and accept operation only when current is 68–75 A and no protection reoperates.

## Expected Result

Successful restart with balanced voltage, current 68–75 A, and no repeat trip.

## Scenario 11: Excessive vibration

### Field Condition

vibration 9.1 mm/s RMS (normal 2–4); axial vibration 6.0 mm/s; bearing 81 °C; speed 1,480 RPM

### Problem

High axial vibration with rising bearing temperature indicates alignment, thrust, or bearing damage.

### What Can Be Done

- If vibration exceeds the applicable OEM/site alarm limit or temperature exceeds 80 °C, unload and stop under the approved procedure.
- Confirm the reading with a second sensor and compare horizontal, vertical, and axial values at both bearings.
- Inspect coupling guard, foundation bolts, pipe supports, and alignment records after LOTO.
- Correct soft foot/pipe strain or replace damaged coupling/bearing components as authorized.
- Conduct a controlled run-up and accept only vibration 2–4 mm/s RMS and bearing temperature 65–75 °C.

## Expected Result

Vibration 2–4 mm/s RMS and bearing 65–75 °C across the operating range.

## Scenario 12: Lubrication problem

### Field Condition

oil level 12 mm below minimum; oil temperature 79 °C (normal 45–60); vibration 6.2 mm/s RMS

### Problem

Low or degraded lubricant is causing friction and heat.

### What Can Be Done

- If oil temperature exceeds 70 °C or level is below minimum, reduce load and stop according to the approved procedure.
- Confirm the oil sight glass and temperature with local instruments; sample oil for water/metal contamination under the maintenance plan.
- Check the breather, cooler, and oil-ring/forced-lube flow; restore a blocked breather or cooler only when safe.
- Drain and refill with the specified lubricant to the marked level; do not mix unapproved grades.
- Restart and verify oil 45–60 °C, vibration 2–4 mm/s RMS, and level between the marked limits.

## Expected Result

Oil level within marks, oil 45–60 °C, vibration 2–4 mm/s RMS.

## Scenario 13: Loss of prime

### Field Condition

discharge 0.4 barg after start (normal 5.0–5.5); flow 0 m<sup>3</sup>/h; suction 1.9 barg; casing level 0 %

### Problem

The casing is not filled or air is entering the suction path.

### What Can Be Done

- Stop the pump if flow remains below 20 m<sup>3</sup>/h for 30 s or if discharge fails to exceed 3.5 barg.
- Verify suction source level, suction valve position, and the casing vent/priming valve using the approved lineup.

3. Inspect the suction flange, mechanical seal, and drain/vent fittings for air ingress without approaching a leak point.
4. Re-prime and vent the casing from a safe location; repair the identified ingress point under isolation.
5. Restart and confirm casing full, discharge 5.0–5.5 barg, and flow 180–200 m<sup>3</sup>/h.

## Expected Result

Casing full, discharge 5.0–5.5 barg, flow 180–200 m<sup>3</sup>/h.

# Scenario 14: Coupling misalignment

## Field Condition

parallel offset 0.42 mm (reference below 0.10); angular error 0.28 mm/100 mm; vibration 7.5 mm/s RMS

## Problem

Alignment is outside the training reference and is producing elevated vibration.

## What Can Be Done

1. Stop and isolate the pump if vibration exceeds the applicable site limit; do not remove the guard while rotating.
2. Confirm alignment with a laser or dial method and record cold/hot offsets; reject readings above 0.10 mm parallel or 0.10 mm/100 mm angular reference.
3. Check soft foot and pipe strain before moving the motor; correct soft foot above 0.05 mm.
4. Re-align the coupling, torque bolts, and reinstall the guard to the approved standard.
5. Run at 1,480 RPM and verify vibration 2–4 mm/s RMS and bearing temperature 65–75 °C.

## Expected Result

Parallel offset below 0.10 mm, angular error below 0.10 mm/100 mm, vibration 2–4 mm/s RMS.

# Scenario 15: Pump overheating

## Field Condition

casing temperature 91 °C (normal 55–70); flow 34 m<sup>3</sup>/h; discharge 5.6 barg; minimum-flow valve closed

## Problem

Low-flow operation is causing internal recirculation and heat buildup.

## What Can Be Done

1. If casing exceeds 80 °C or flow is below 50 m<sup>3</sup>/h, open the approved minimum-flow path or stop the pump.
2. Confirm the minimum-flow valve position and verify the return line is not blocked using the pressure indicators.
3. Check that discharge control is not forcing the pump against a closed valve; restore flow above 150 m<sup>3</sup>/h where process conditions permit.
4. Inspect impeller and casing for rub damage after cooling, isolation, and depressurization.
5. Restart and confirm casing 55–70 °C, flow 180–200 m<sup>3</sup>/h, and discharge 5.0–5.5 barg.

## Expected Result

Casing 55–70 °C and flow 180–200 m<sup>3</sup>/h with the minimum-flow path available.